TRACE-ES framework for the Global South

Translating Regional Analytics into Credible, Equitable ES evidence

Drivers: climate and environmental change

- Warming (SSP–RCP)
- Land-use change
- Urban, cropland
- Sea-level rise
- Drought, aridity
- Restoration, zoning

alter ES supply and demand

Evidence from this review (1990–2025, n = 7,972)

Gains

- ≈25% growth per year; doubling 3.2 yr
- Land-change × climate scenarios
- New services: cooling, flood, marine

Gaps

- Validation reported in <10% of studies
- 25–38% of regional studies led abroad
- 48.9% China; South–South links 8.0%

gaps motivate four pillars

T Transferable quantification

- Tier 1: coefficients, screening only
- Tier 2: InVEST, SWAT, RUSLE
- Tier 3: process–ML hybrids
- Regional parameter libraries

R Robustness & credibility

- Ensembles; uncertainty propagation
- In-situ validation, field data
- Model cards and reporting
- Context-sensitive ESV coefficients

A Analytical integration

- Land change (PLUS/FLUS) × climate
- ES supply–demand flows
- Trade-offs and synergies
- Attribution (Geodetector, MGWR)

C/E Co-production & equity

- Global South-led teams
- Data sovereignty (CARE)
- Participatory, cultural NCP
- South–South hubs; open outlets

Deliverables

- **T: Calibrated models**
- **R: Validated estimates**
- **A: Planning-ready scenarios**
- **C/E: Locally owned evidence**

Outcomes 2026–2035: decision-grade ES intelligence

- Ecosystem accounts (SEEA-EA)
- Nature-based solutions targeting
- Food–water–carbon planning
- GBF and SDG 2, 6, 11, 13–15